

Design of High-Efficiency RF Class-D Power Amplifier

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I. INTRODUCTION

CLASS-D RF power amplifiers have been in use for a long time. Because of the switching nature of this amplifier, it has achieved efficiencies of 80% at low power (10 to 100 W) and of 70% at higher power (100 to 500 W). However, the theoretical efficiency of a Class-D amplifier is 100%. The reason for the lower efficiency in practice has been the increased switching losses associated with the RF operation. Early papers on Class-D amplifiers neglected the effect of these switching losses [1], [2]. In this paper, the losses in an rf switching power amplifier and their frequency dependence are described. The losses analyzed are the switching, conduction, and gate drive losses. The last of these are usually ignored in lower frequency applications, but must be accounted for at the frequencies of interest in this paper (> 10 MHz). Expressions are derived to predict the efficiency of a Class-D power amplifier.

In recent years, Class-D circuits operating with zero-voltage switching using the dead time have been introduced [3]–[5]. However, none of these papers describe the design equations or give the conditions for zero-voltage switching. At rf the available dead time is very short therefore the design space is more restricted. In this paper, the circuit is analyzed to predict the proper magnitude of the dead time and the load current to allow zero-voltage switching of the devices. There are also issues related to the control of that dead-time with a reasonable accuracy when the dead-time is in nanoseconds. Using a square wave gate drive to do the timing quickly becomes very difficult as the switching frequency gets to the megahertz range and above. A method whereby the amplitude of the sinusoidal gate voltage controls the dead time is described [6]. The design equations for the zero-voltage switching Class-D circuit are derived.

A 300-W, 13.56-MHz, Class-D circuit is designed in the traditional manner to illustrate the magnitude of the different types of loss. A circuit using the ZVS equations developed in this paper is designed. An experimental circuit is built using standard IRF540 devices in TO220 packages. That circuit does not meet its performance goals because of the package inductance. A new low inductance half-bridge package is introduced to solve this problem. Techniques for circuit layout and power measurements for rf applications are also presented in the experimental section. A low loss gate drive circuit is also

presented using a Class-E circuit to provide the drive power. The experimental results confirm the accuracy of the design equations derived in this paper.

II. LOSSES IN SWITCHING POWER AMPLIFIERS

In contrast to the traditional Class A, B, and C amplifiers, switched-mode power amplifiers such as the Class-D amplifier have a theoretical efficiency of 100%. Thus, if one assumes that the switching devices are “ideal” (i.e., have zero resistance when on, infinite resistance when off, no associated parasitic capacitance or inductance, and zero transition times), then the losses in the amplifier are zero. In the other amplifiers, inherent losses caused by the linear mode operation of the devices make the theoretical efficiency much less than 100%. However, such ideal devices and components do not exist, so one must contend with some losses in the switching power amplifiers. In this section, the losses in switching power amplifiers are discussed with emphasis on the effect of operation at radio frequencies.

A. Conduction Losses

Conduction losses include all the power dissipation caused by the resistances associated with the semiconductor devices, the inductors, and the capacitors. The conduction losses in the semiconductor are not affected by the operation at higher frequencies, primarily because the skin depth in semiconductors at the frequencies of interest are large relative to the size of the device.

Therefore, the operating frequency has no direct impact on conduction losses within the semiconductor devices. However, the skin effect adversely affects the losses in inductors, capacitors, and in all conductors carrying large rf currents. At the frequencies of interest, multilayered ceramic capacitors exist that exhibit very low loss at frequencies up to hundreds of megahertz. Similarly, proper design of inductors and of all connections that carry large currents can reduce the losses resulting from the skin effect. In general, one should use wide, flat, and short connections to reduce losses as well as to reduce the effect of parasitic lead inductances.

B. Turn-On Switching Losses

At the turn-on instant, the device has a high voltage across its output terminals and no current. The voltage starts to fall from its initial value and the current starts to rise toward its final value. If this voltage fall is not instantaneous, there is a crossover period when both the current and the voltage are

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nonzero. A significant amount of energy can be dissipated during this time [7], [8]. The power dissipation due to this mechanism is proportional to the length of the crossover period. It is also proportional to the switching frequency as the energy loss occurs every switching cycle. This loss mechanism is negligible in a Class-D circuit as the turn-on occurs essentially at a zero-current crossing.

A more important loss mechanism that occurs at the turn-on transition is due to the discharge of the energy stored in the output capacitor of the semiconductor power devices. This capacitance is intrinsic to the device, a result of the existence of PN junctions. For example in a MOSFET, the output capacitance, C_{oss} , is formed by the drain-to-drain capacitances. Even in an idealized system—that is, one with zero resistance and zero switching times—this discharge loss cannot be eliminated.

The device capacitances are nonlinear and vary with the drain-to-source voltage. In order to simplify the analysis, the nonlinearity of the device capacitances is ignored in this paper. The capacitance is assumed to be fixed and the value of C_{oss} at a drain-to-source voltage of 25 V is used. The effect of the nonlinearity can be accounted for and will be the topic of a future paper. In a Class-D circuit, (Fig. 3), there are two devices that switch on and off alternately. If the first device is off, then the voltage across it is equal to the rail voltage V_{dc} and the energy stored in the output capacitor C_{oss} is

$$E_d = 1/2 C_{oss} V_{dc}^2. \quad (1)$$

This energy is being dissipated once in every cycle of the switching frequency; therefore, the discharge power loss is

$$P_d = 1/2 C_{oss} V_{dc}^2 f_s. \quad (2)$$

Simultaneously, the capacitor of the second device is charged through the resistance of the first device to the rail voltage. As we assume the output capacitances are fixed, then the charging process dissipates an amount of energy equal to the amount of energy being stored. Thus, the turn-on loss in a Class-D circuit at every switching transition will be twice the loss in (2). As there are two switching transitions per cycle, the total turn-on losses in a Class-D circuit is four times the loss in (2). This loss mechanism can easily be the dominant one, especially in cases where the voltage across the device is relatively large, i.e., 50 V or more. A calculation of the turn-on switching losses is presented for a practical circuit in Section III. As mentioned above, the output capacitances of the power devices are intrinsic in the structure of the devices, and cannot be reduced to zero. Thus, the only practical means for reducing this power loss would be to achieve zero-voltage switching.

C. Turnoff Switching Losses

The same type of crossover power loss that occurs at turn-on also occurs at turnoff. In contrast to the turn-on period, there is no capacitor voltage discharge loss. However, there is a dual type of loss caused by the current flowing in parasitic inductors. At turnoff, the current is flowing through the leads and wire bonds into the device. In general, parasitic

inductances are associated with these leads. The amount of energy stored in these inductances is equal to

$$E_L = 1/2 L I^2 \quad (3)$$

where L is the parasitic inductance value and I is the current flowing at the turnoff instant. The power loss from this energy discharge is also proportional to the switching frequency and is given by

$$P_L = 1/2 L I^2 f_s. \quad (4)$$

In contrast to the capacitive discharge losses, in which the device capacitances are not under the control of the circuit designer, the inductive discharge losses can be reduced by proper design of the circuit layout or by using low lead inductance packages.

D. Gate Drive Losses

At low frequencies, MOSFET's consume very little gate drive power, which is one of their most interesting features. However, as the frequency increases, the power dissipation caused by charging and discharging the gate capacitance becomes significant. For the sake of simplifying the analysis, the gate circuit is represented by a series RC circuit [Fig. 1(a)]. Because of the Miller feedback effect of the gate-drain capacitance and the variation of the device capacitances with voltage, the gate charge required to turn on a device is the best way to correctly predict the gate drive requirements [9]. The gate drive power requirement is dependent on the type of drive being used. We will examine two types of gate drives: a square-wave gate drive and a sinusoidal resonant gate drive. The two drive schemes will have different losses for a given peak gate voltage and switching speed. The peak current requirements of each system are also different.

In the case of a square-wave drive, the input voltage is a square wave but the input current is a pair of pulses [Fig. 1(b)]. The total charge required by the gate (given by the hatched area under the gate current in Fig. 1(b) and the peak voltage across the gate determines the amount of energy stored in the gate capacitance as follows:

$$E_{gate} = 1/2 V_{gs} Q_g \quad (5)$$

where Q_g is the charge required to raise the gate voltage to V_{gs} . Note that Q_g is obtained from the device data sheets and that it includes the amount of charge stored in the gate-drain capacitor as well as the gate-source capacitor. As the gate is being charged through a resistor, the same amount of energy is dissipated in that resistor during the charging process. Similarly, when the gate is discharged, the stored energy is dissipated resistively. Therefore, the loss associated with turning the device on and off per cycle is two times E_{gate} , and the power loss is given by

$$P_g = V_{gs} Q_g f_s. \quad (6)$$

The peak current required to switch an MOSFET is dependent on the desired switching speed. The time constant τ of

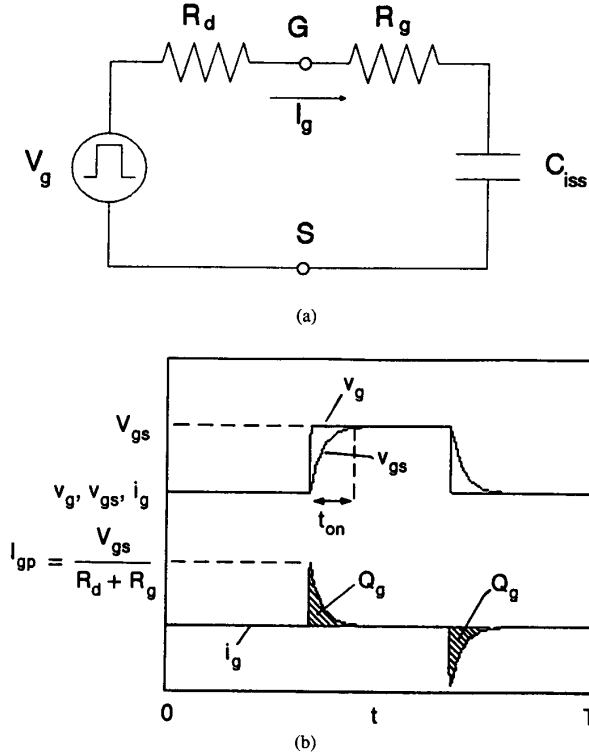


Fig. 1. (a) The simplified equivalent circuit of the gate and gate drive circuit. (b) The input voltage waveform, v_g , the voltage across the gate capacitor, v_{gs} , and the input current waveform, i_g .

the RC network determines the length of time t_{on} it takes the gate capacitance to charge up to V_{gs} . Typically, it takes a total time equal to 4τ to fully charge a capacitor. Therefore, for a gate capacitance C_{iss} , one can calculate a desired value of gate resistance R_g and gate drive circuit resistance R_d as follows:

$$t_{on} = 4\tau = 4(R_d + R_s)C_{iss} \quad (7)$$

The value of C_{iss} used here is the value at $V_{ds} = 0$, which gives a worst-case scenario for the gate and drive resistances. The peak current in the gate drive circuit is then given by

$$I_{gp} = \frac{V_{gs}}{R_d + R_g} \quad (8)$$

Note that the power requirement for the gate drive with a square-wave voltage is independent of the switching speed, and the switching speed is ultimately limited by the gate resistance. Thus, there is an intrinsic limit to how fast the gate can be charged, which becomes a factor as the frequency of operation increases.

The sinusoidal resonant gate drive requirements and power losses require a different calculation (Fig. 2). The input voltage waveform is sinusoidal and if one ignores the nonlinearity of the input gate capacitance, the current is also sinusoidal. Thus, the peak current needed to charge the gate from zero to V_{gs} in a time t_{on} , with a sinusoidal gate drive of frequency f_x can

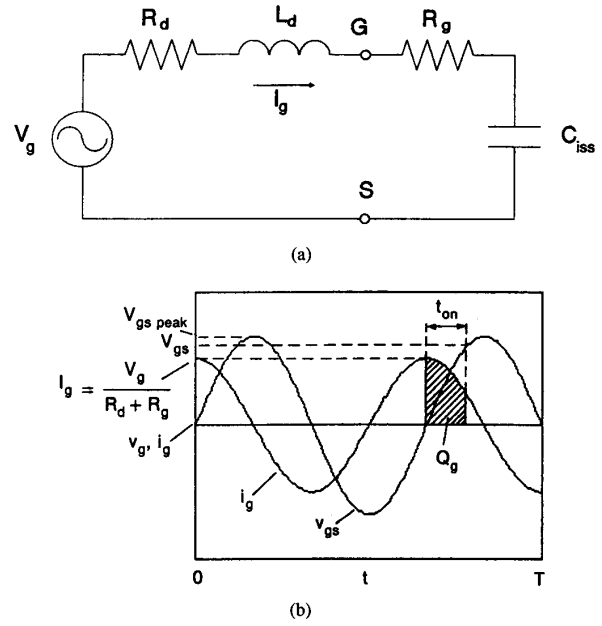


Fig. 2. (a) The simplified equivalent circuit of a resonant gate drive circuit. (b) The input current and the voltage across the gate capacitor.

be calculated from the required gate charge Q_g as follows:

$$I_g = \frac{Q_g}{\int_0^{t_{on}} \cos(\omega_s t) dt} \quad (9)$$

$$= \frac{Q_g \omega_s}{\sin(\omega_s t_{on})} \quad (10)$$

Note that V_{gs} is only equal to the peak voltage of the sine wave when t_{on} is equal to a quarter of the switching time t_s . The peak gate current under this condition is then equal to

$$I_g = Q_g \omega_s \quad (11)$$

The peak voltage capability of the gates of the MOSFET will then clearly limit how fast one can switch the device with a resonant sinusoidal gate drive. The peak voltage across the gate of the device is given by

$$V_{gs_peak} = \frac{V_{gs}}{\sin(\omega_s t_{on})} \quad (12)$$

Because the gate capacitance is being resonated by the resonant inductor L_d in Fig. 2(a), the reactive impedance of the gate circuit is considered negligible. Thus, the required input voltage is equal to

$$V_g = I_g(R_d + R_g) \quad (13)$$

and the power dissipation is

$$P_g = 1/2 I_g^2 (R_d + R_g) \quad (14)$$

The above equations yield approximate values because the nonlinearity of the device capacitance and the Miller effect have not been taken into account. In contrast to the square-wave gate drive, the sinusoidal resonant gate drive is not

limited in switching speed by the gate resistance, yet the power requirement is proportional to the resistances. One can reduce the time it takes to reach a certain gate voltage by increasing V_g to increase the peak current I_g , subject to the condition that the peak voltage across the gate capacitance does not exceed its rated capability. The power losses in the gate drive can be reduced by reducing the drive resistance. This is because in a resonant configuration, the energy stored in the gate gets stored for a half cycle in the resonant inductor and is recovered during the next half cycle, but with square-wave switching, the energy is always resistively dissipated.

In general, the square-wave gate drive has much more severe requirements than the resonant gate drive. The gate drive devices must be very fast, and must be capable of sourcing or sinking large currents. For example, to charge a 2000-pF capacitor in 10 ns to 10 V, total gate drive resistance must be equal to 1.25Ω ; thus, the peak current required is 8 A. In order to compare a square-wave gate drive with a sinusoidal gate drive, we need to assume an operating frequency because the required peak gate current increases with frequency. The maximum current in a sinusoidal drive scheme would then occur at the highest operating frequency. The highest operating frequency, for a given switching time, is when that switching time is equal to a quarter of the full cycle. Otherwise, the device never reaches either the fully on or the fully off condition. Thus, the maximum operating frequency given a 10-ns switching time is 25 MHz. Under this worst-case condition, the peak gate drive current required is 3.14 A for the aforementioned example. Thus, the peak current requirement of the sinusoidal resonant gate drive is always less than that of the square-wave gate drive for a given switching time. Another advantage of the sinusoidal gate drive is that the design of the gate drive transformers is much simpler. In a sinusoidal gate drive, the transformer is a narrowband transformer; the square-wave drive requires a broadband transformer. Similarly, all lead inductances and leakage inductances in the gate drive circuit can be absorbed into the resonant inductor L_d , but they have to be minimized for the square-wave drive.

III. TRADITIONAL DESIGN OF CLASS-D POWER AMPLIFIER

A voltage-switching Class-D circuit consists of a pair of power devices in a cascade connection (Fig. 3). The devices are switched on and off alternately. The input dc voltage is connected to the drain of the top device and the source of the lower device is connected to ground. The devices are driven via a three-winding transformer with appropriate polarities on the output so that the same drive is used for both devices. The midpoint between the two devices is connected to the load circuit. In this discussion, the load network is assumed to be a series RLC network. The traditional design assumes that the RLC resonant frequency is the switching frequency of the devices [1]. Therefore, because the devices switch alternately, the voltage at the midpoint (v_c) is a square wave voltage of amplitude equal to V_{dc} , if the devices are assumed to be ideal.

Since the load network is a tuned circuit that provides little impedance to the fundamental component of the voltage v_c and high impedance to the higher harmonics, one can assume,

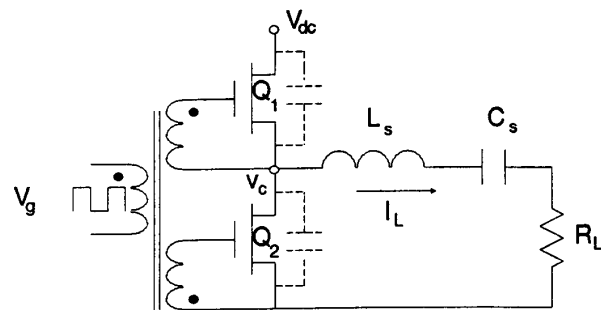


Fig. 3. Voltage-switching Class-D power amplifier.

with a reasonable $Q(\sim 10)$ that the current flowing is the one driven by the fundamental component of the input voltage. Because the voltage is a square wave, its Fourier expansion is given by

$$v_c(\omega t) = V_{dc} \left[\frac{1}{2} + \frac{2}{\pi} \sin(\omega t) + \frac{2}{3\pi} \sin(3\omega t) \cdots \right]. \quad (15)$$

As the impedance of the RLC series load at resonance is equal to R_L , the current is given by

$$i_L(\omega t) = \frac{2V_{dc}}{\pi R_L} \sin(\omega t). \quad (16)$$

Each of the devices carries the current during one half of the switching cycle, so (16) determines the peak device current. The output power is then given by

$$P_0 = \frac{2}{\pi^2} \frac{V_{dc}^2}{R_L}. \quad (17)$$

We will design a 300-W Class-D amplifier operating at 13.56 MHz from a 75-V dc input voltage. From the design equations (16) and (17), we get the following values for the load resistance and the device peak current:

$$R_L = 3.8 \Omega \quad I_L = 12.56 \text{ A}.$$

A suitable MOSFET for the above design is the IRF 540, which is a 100-V, 28-A device. The efficiency of that Class-D design can be predicted using the expressions for the different types of losses from Section II. Let us assume for the sake of simplicity that the inductors and capacitors in the circuit are ideal and do not contribute to the losses.

The conduction losses are therefore the losses in the devices. The IRF 540 has an on-resistance, R_{dson} , equal to 0.085Ω . Since each device carries the load current for half a cycle, the conduction losses per device are

$$P_{cond} = 1/2 \left(\frac{I_L}{\sqrt{2}} \right)^2 R_{dson} = 3.35 \text{ W}. \quad (18)$$

The devices are switching on and off at zero current, which reduces the inductive discharge losses and the crossover losses to zero. The only switching losses that remain are the capacitive discharge losses. As mentioned in Section II-B, the losses

per device are equal to twice the losses given by (2). The output capacitance of the device is equal to 500 pF (at $V_{ds} = 25$ V), thus the discharge losses are

$$P_{dis} = 38.14 \text{ W}. \quad (19)$$

The gate charge required to turn on an IRF 540 is 39 nC for a gate voltage of 10 V, so from (6), the gate drive losses with a square-wave drive are

$$P_g = 5.3 \text{ W}. \quad (20)$$

Therefore, the overall efficiency for the traditional design is

$$\eta = \frac{P_{out}}{P_{out} + 2P_{dis} + 2P_{cond} + 2P_g} = \frac{300}{300 + 2(38.14) + 2(3.35) + 2(5.3)} = 0.76.$$

Clearly, the losses are dominated by the switching losses in the above design. These switching losses are usually neglected in the traditional analysis, and only the conduction losses are taken into account [1]. In practice, one can expect even worse performance as the devices, each of which is dissipating 47 W, will operate at a high junction temperature and the on-resistance would go up by as much as 70 to 80% [10]. However, in practice the measured efficiency is better than the predicted efficiency [11]. In fact, as we will show in the next section, there is a lossless discharge mechanism in Class-D circuits (caused by the nonideal switching of the devices) that in general improves the efficiency of actual circuits. This lossless discharge mechanism can be used in Class-D circuits to achieve ideal zero-voltage switching and thus totally eliminate the discharge losses [4], [5].

IV. ZERO-VOLTAGE SWITCHING CLASS-D POWER AMPLIFIER

To explain the higher efficiencies achieved in practice in Class-D circuits the switching transition between the devices needs to be examined carefully. Fig. 4 shows the four different switching networks that occur during a switching cycle and the midpoint voltage and load current during those periods. The previous analysis has assumed an instantaneous transition between the devices. However, in practice, a dead time occurs during the period when one device has turned off before the other has turned on.

Before the switching instant, one device is on and is conducting the load current in the direction shown in Fig. 4(a) and the midpoint voltage is equal to V_{dc} . (Note: to simplify the analysis, we shall assume that the forward drop of the devices is negligible.) The top device is then turned off while the lower device is still off. During this transition period (or dead time), if the resonant load circuit phase is lagging, the current continues to flow through the capacitances of the devices [Fig. 4(b)]. The current discharges the capacitor of one device while it charges the capacitor of the other device. If one assumes that the output capacitances of the devices are equal then the current is equally divided between them. At the end of the transition period, the second device turns on, the midpoint voltage goes to zero, and the current reverses direction in the load [Fig. 4(c)].

At the end of this on-period, the lower device turns off and the current flow in the capacitors, charging the lower one and discharging the upper one [Fig. 4(d)]. In traditional circuits, a lossless partial discharge of the capacitor occurs that reduces the energy in the capacitor at the turn-on instant, which reduces the switching loss. Hence, a finite-time switching transition enables the circuits to achieve a higher efficiency than expected.

A. Analysis of Zero-Voltage Switching in Class-D Power Amplifier

It is possible to control the dead time, the peak current in the devices, and the load phase angle relative to the fundamental component of the voltage to achieve zero-voltage switching and to eliminate the switching losses totally. The following analysis refers to the waveforms in Fig. 5, which have a slightly shifted time origin to simplify the analysis. During the transition period the voltage at the midpoint is given by

$$v_c(t) = \frac{1}{2C_{oss}} \int_{(t_s/4)-(t_d/2)}^t I_L \cos(\omega_s t - \phi) dt. \quad (21)$$

The midpoint voltage reaches its minima at the zero-current crossing. To achieve zero-voltage switching, the end of the dead time should coincide with the current voltage crossing and the amplitude of the current should be sufficient to make the voltage minima equal to zero. Therefore, the zero-voltage switching condition can be written as

$$V_c = \frac{1}{2C_{oss}} \int_{(t_s/4)-(t_d/2)}^{(t_s/4)+(t_d/2)} I_L \cos(\omega_s t - \phi) dt \quad (22)$$

where ϕ is equal to

$$\phi = \frac{t_d}{t_s} \pi. \quad (23)$$

The integral is solved to give

$$V_c = \frac{I_L}{2\omega_s C_{oss}} [1 - \cos(2\phi)]. \quad (24)$$

Before proceeding with the analysis it is important to see what happens if the above condition is unsatisfied. If the dead time is too short [Fig. 6(a)] or too long [Fig. 6(b)], the voltage will not be at zero and a certain loss would occur. Similarly, if the current amplitude is too low to discharge the voltage, some discharge loss results [Fig. 6(c)]. On the other hand, if the current is too large, the voltage crosses zero twice [Fig. 6(d)] and reverses polarity across the device. This situation is potentially destructive because the negative voltage forward biases the MOSFET body diode and turns it on. As the reverse recovery time of the body diode is of the order of 100 ns, it will short out the other device when it turns on and the device will fail. It should be pointed out that the optimum operating point is broad compared with the peak current value. The current level needed to make the voltage go negative is so high that the device will probably fail because of too high of a conduction loss long before the diode failure occurs. However, it is important to keep the different failure mechanisms in mind when designing an actual circuit.

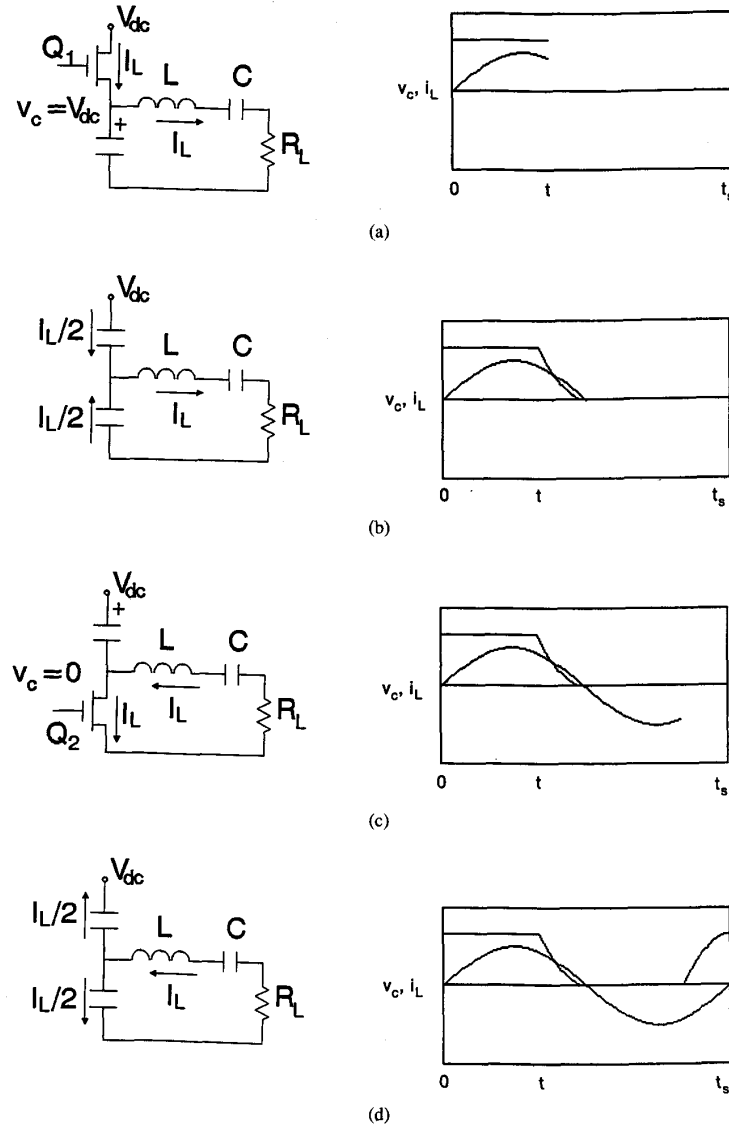


Fig. 4. The four switching networks that appear during a full switching cycle in a Class-D circuit. (a) The top device, Q_1 , is conducting, and the bottom one, Q_2 , is off. (b) The first transition period occurs when Q_1 is turned off while Q_2 remains off. (c) Q_2 is turned on while Q_1 remains off. (d) The same as (b), except that Q_2 is turned off and the current flow is reversed in the capacitor.

For the purpose of calculating the power output, it is simpler to assume that the voltage at the midpoint between the devices is trapezoidal. In reality, the voltage across the capacitors falls sinusoidally under the action of the resonant load current during the dead time, as described by (21). However, the difference between the fundamental components in both cases is relatively small. The angle ϕ , for a trapezoidal voltage waveform, is the phase angle between the current and the fundamental component of the voltage waveform. The midpoint voltage is then equal to

$$V_{cf} = \frac{4}{T_s} \int_0^{T_s/2} v(t) \cos(\omega_s t) dt \quad (25)$$

$$= 2 \frac{V_{dc}}{\pi \phi} \sin(\phi). \quad (26)$$

The power delivered to the load is then given by

$$P_0 = 1/2 V_{cf} I_L \cos(\phi). \quad (27)$$

Substituting for V_{cf} from (26) in (27) yields

$$P_0 = \frac{V_{dc}}{\pi \phi} I_L \sin(\phi) \cos(\phi) \quad (28)$$

$$= 1/2 \frac{V_{dc}}{\pi \phi} I_L \sin(2\phi). \quad (29)$$

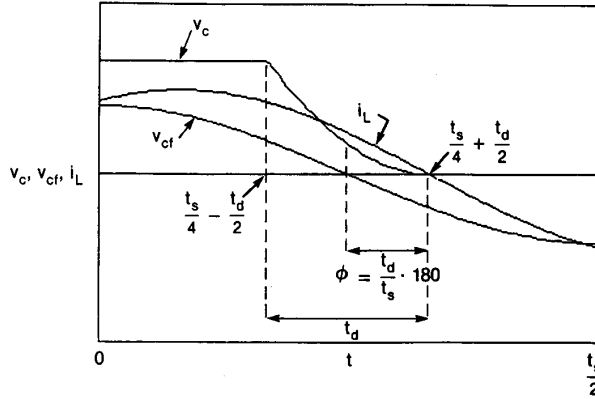


Fig. 5. Waveforms of the midpoint voltage v_c , the fundamental component of the midpoint voltage v_{cf} , and the load current i_L .

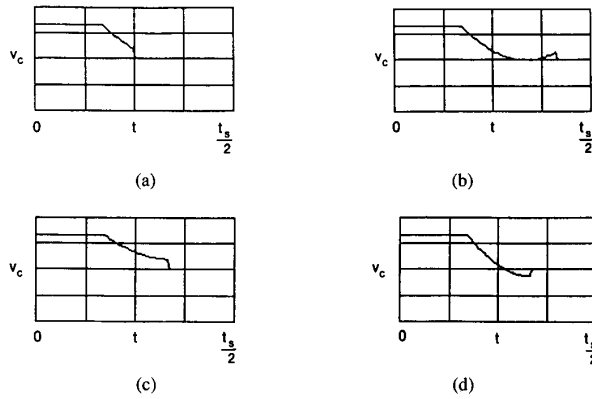


Fig. 6. Four possible nonoptimal switching situations. (a) The dead time is too short. (b) The dead time is too long. (c) The current is too low to fully discharge the capacitance. (d) The current is too high, which causes the voltage across the device to switch polarity.

Substituting the value of V_{dc} that satisfies the zero voltage condition (24) into (29) yields the equation for the load current:

$$I_L = \sqrt{\frac{4P_0\pi\omega_s C_{oss}\phi}{\sin(2\phi)[1 - \cos(2\phi)]}}. \quad (30)$$

Equations (30) and (24) give the required dc input voltage and the load current for a given power output, frequency of operation, dead time, and device output capacitance for zero-voltage switching of the devices. Therefore, given the output power and the frequency of operation, one can generate a family of VI curves, parameterized by C_{oss} . Each point on these curves satisfies the power output requirement under optimum, i.e., zero-voltage, switching condition at a specific dead time.

Fig. 7 shows such a plot of the VI curves for two commercial devices. By superimposing the safe operating area (SOA) of the devices on the curves, one can determine whether or not a particular device is suited for the particular design conditions. As can be seen, the IRF 540 is suitable for the 300-W circuit at 13.56 MHz. The value of the dead time remains to be determined. This can be done graphically by either plotting the

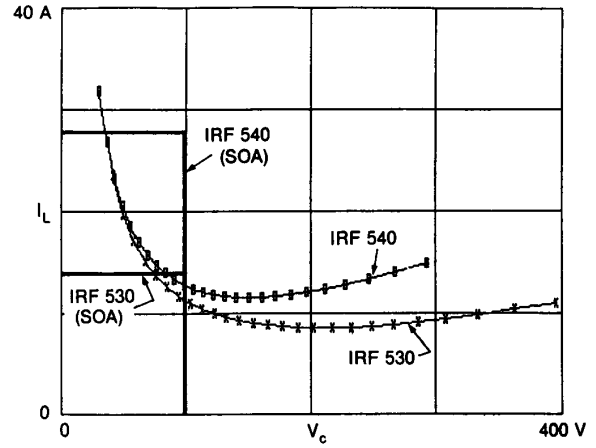


Fig. 7. VI curves satisfying the zero-voltage switching condition for a power output of 3000 W at 13.56 MHz for the IRF 540 (boxes) and the IRF 530 (crosses) power MOSFET's. The safe operating areas (SOA) of the two devices are superimposed on the graph (thick lines).

input dc voltages versus the dead time or the current versus the dead time (Fig. 8). In the case of the IRF 540 devices, a suitable dead time is 12 ns, which places the operating point at about 72 V and 15 A. This is a reasonable distance from both the maximum current and voltage ratings of the device.

The control of the dead time is a matter of practical importance. At these frequencies, precisely controlled square waves are difficult to generate, and to control the dead time to an accuracy of 1 ns could be very expensive. As mentioned earlier, a square-wave gate drive scheme would require the design of a well-matched broadband system, which is much more difficult to do than designing a single frequency matched system. In the next section, a sinusoidal drive scheme in which the dead time is controlled by the amplitude of the gate voltage is introduced and some of its limitations are discussed.

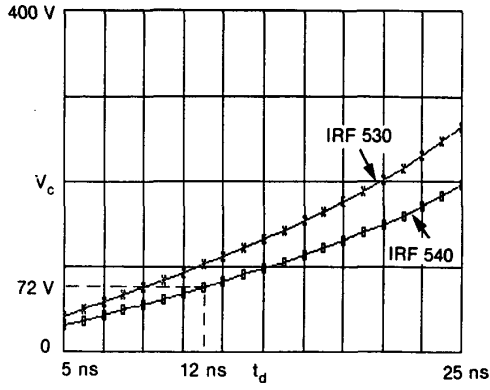
B. Sinusoidal Gate Drive With Dead-Time Control

MOSFET's have a threshold gate voltage level below which they are off. In the case of power switching MOSFET's, such as the International Rectifier Hexfets, the transition from off to on occurs very fast once the threshold is crossed. This property of the devices allows us to use a sinusoidal input voltage to control the dead time between the two devices. The gate voltages are sinusoids out of phase by 180° (Fig. 9). The gate voltage of one device will cross the threshold to turn that device off while the gate voltage across the other device is still below threshold. Thus, during this crossover period the two devices are off, which is the desired state. Knowing the threshold voltage of the devices, one can determine the gate drive voltage required for a specific dead time.

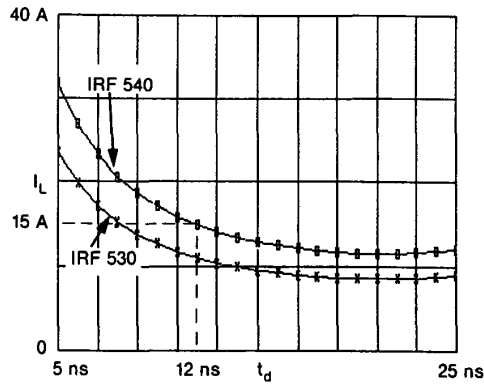
The voltage across the gate source capacitance is

$$V_{gs}(t) = V_{gs} \sin(\omega_s t). \quad (31)$$

The gate voltage is equal to the threshold voltage V_t at time



(a)



(b)

Fig. 8. (a) The curve of input dc voltage versus dead time for the same devices as in Fig. 7. (b) The curve of device peak current versus dead time for the same devices. The chosen operating point for the IRF 540 is also shown.

$t_d/2$, which gives

$$V_t = V_{gs} \sin \left(\omega_s \frac{t_d}{2} \right) \quad (32)$$

, which can be rewritten as

$$V_{gs} = \frac{V_t}{\sin \left(\omega_s \frac{t_d}{2} \right)}. \quad (33)$$

Thus (33) gives the required gate voltage for a given dead time. However, it is important to first examine the range of dead times achievable using this gate drive scheme. In order to do this, it is easier to rearrange (33) as follows:

$$t_d = \frac{2}{\omega_s} \sin^{-1} \left(\frac{V_t}{V_{gs}} \right). \quad (34)$$

The first limitation is that the peak gate voltage of the devices is usually limited to 20 V. This sets a limit to how short the dead time can be

$$t_{dmin} = \frac{2}{\omega_s} \sin^{-1} \left(\frac{V_t}{20} \right). \quad (35)$$

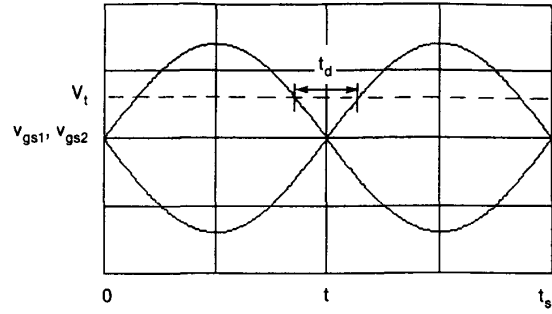


Fig. 9. Two sinusoids out of phase by 180° , showing the threshold voltage, V_t , of the device and the resulting dead time.

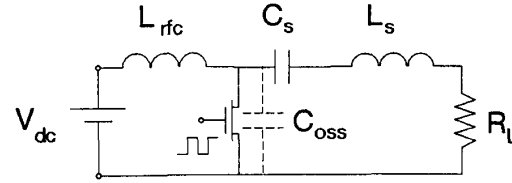


Fig. 10. Circuit diagram of Class-E zero-voltage switching circuit.

At 13.56 MHz and assuming $V_t = 3.5$ V, the minimum dead time is equal to 4.1 ns. The other limit to the dead time occurs when the gate source voltage becomes too low to turn on the device properly. As a matter of good design practice the gate voltage should not be lower than about 6.5 V. Thus, the largest dead time one can achieve is slightly higher than 13 ns. The dependence of the dead time on the amplitude of the gate drive voltage makes the control circuit design very simple and economical. However, it does limit the range of dead times over which one can operate the circuit.

The dead time chosen for the circuit design with the IRF 540 was 12 ns. This requires a gate source voltage of 7.2 V. From the data sheets for the IRF 540, the required gate charge for the above gate voltage is 30 nC. From (11) the gate current is then equal to 26 A. The gate resistance of the IRF 540 is equal to 1Ω . Thus, the drive power requirement for this design is equal to 3.27 W per device. The total gate drive requirement, however, must include the efficiency of the gate drive circuit. The gate drive losses are not an insignificant portion of the total losses. The value of the gate resistance is beyond the circuit designer's control, so the best one can do is to generate that gate drive as efficiently as possible. Fortunately, high-efficiency resonant circuits exist to generate this gate drive. The simplest circuit (because it uses a single transistor) is a Class-E circuit (Fig. 10) [12]. It is also possible to use a Class-D circuit whose two devices would have much smaller drive requirements so they can be driven directly from the crystal oscillator circuit.

The design equations for the Class-E circuit have been derived for different operating conditions in [13]–[15]. These can be used to design the driver for the Class-D circuit, keeping in mind that the output in the driver case is not the power to the gate resistance but the voltage across the

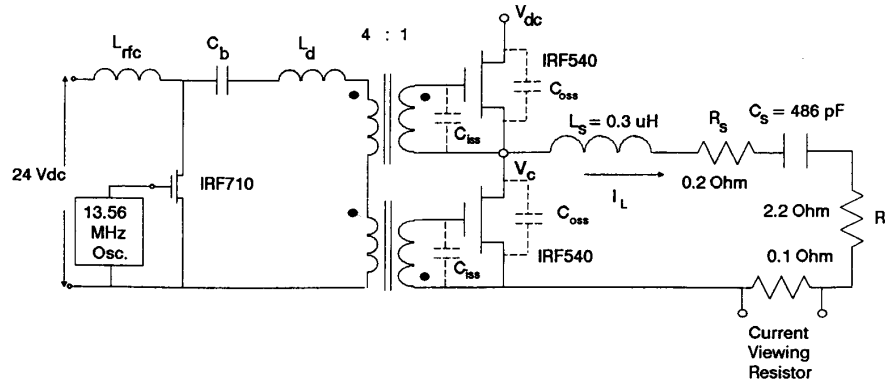


Fig. 11. Circuit diagram of experimental Class-D 300-W power amplifier.

gate capacitance. One other consideration is the transformer connection. The two gates could be driven from a single transformer, with two secondaries connected with opposite polarities to give the 180° phase shift between the two gate voltages. This has the obvious advantage of using only one gate drive transformer. However, this advantage may be fully offset by the need to eliminate a positive feedback mechanism that results in increased losses. The ringing in the drain circuit of the conducting device is coupled into the gate of the other device by way of the Miller feedback capacitor and the gate drive transformer windings. The gate drive modulation increases the amplitude of the ringing, which completes the feedback loop. The increased ringing implies large circulating currents in the devices that cause additional conductive losses. Therefore, it may be preferable to drive the gates through two transformers with series-connected primaries or with parallel-connected primaries. The choice would affect the reflected load into the primary and it is the reflected load that needs to be used in the Class-E circuit design. The detailed design of the gate drive circuit is beyond the scope of this paper and therefore will not be given here.

V. EXPERIMENTAL VERIFICATION OF RESULTS

An experimental circuit based on the design equations given in the preceding section was built and tested (Fig. 11). The design point chosen for the IRF 540 with an 11.5 ns dead time gives an input voltage of 72.5 V and a load current of 15.3 A peak. The load phase angle is $\phi = 28^\circ$ and the load resistance for 300 W is then equal to

$$R_L = \frac{2P_0}{I_L^2} = 2.55 \Omega. \quad (36)$$

The load-phase angle is used to give the value of the load reactance

$$X_L = \omega_s L_s - \frac{1}{\omega_s C_s} = R_L \tan(\phi). \quad (37)$$

A reasonable value of Q for the load circuit is 10. Therefore, the series inductor has to be equal to

$$L_s = Q \frac{R_L}{\omega_s} = 0.3 \mu\text{H}. \quad (38)$$

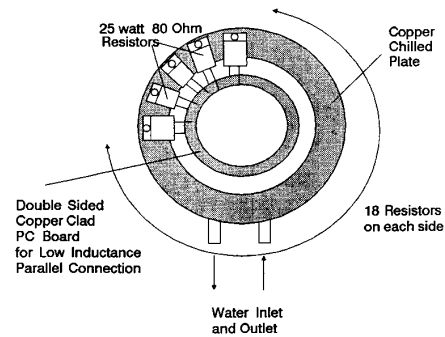


Fig. 12. The load resistance used for calorimetry.

The required series capacitance needed to provide the proper phase angle can now be calculated from equation 37:

$$C_s = 486 \text{ pF}. \quad (39)$$

The load resistance was obtained by paralleling 36 $80\text{-}\Omega$ 25-W resistors in TO3 packages. These resistors were mounted on two annular copper chilled plates welded to a $1/8$ in. copper tube for water cooling (Fig. 12). The resistors are all connected in parallel using a smaller ring made of double-sided printed circuit board. Each side of the board forms one terminal of the load resistor. The whole assembly is placed in a thermally insulating enclosure with openings for the water inlet and outlet and for the electrical terminals. Other circuit resistances, such as the series resistance of the inductor (0.2Ω) and the current viewing resistor (0.1Ω), provided the additional resistance needed to reach 2.5Ω . The water inlet and outlet temperatures were measured with thermocouples to provide a calorimetric measurement of the power delivered to the loads. The calorimeter was calibrated at dc at a fixed rate of water flow to give the dissipated power in the load versus the differential voltage on the inlet and outlet thermocouples.

The series capacitance was provided by the parallel combination of four 100-pF Murata Erie capacitors and a 0- to 100-pF variable multiple plate air capacitor. The Murata capacitors are glass-encapsulated multilayered ceramic capacitors with an

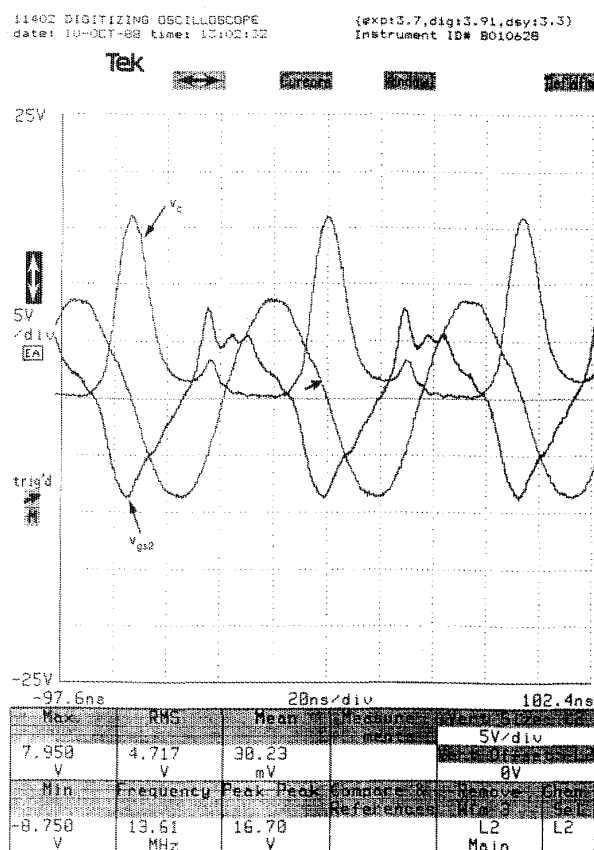


Fig. 13. The midpoint voltage waveform of the circuit built with standard TO-3 packaged devices shows the ringing at 30 MHz.

rf rms current rating of 8 A each. The series inductor was a combination of parasitic inductances of the resistors and an air core inductor consisting of four turns of 1/8 in. copper tubing with an inner diameter of 1 in. The total series inductance was measured to be close to $0.4 \mu\text{H}$, including parasitic inductances on the board and in the resistors. The series capacitance was then changed to maintain the proper load phase angle. The series resistance of the coil was measured on an HP 4324A Q-Meter at 13.56 MHz, and was found to be equal to 0.15Ω at room temperature and 0.2Ω at the operating temperature.

The first attempt at testing the circuit used devices packaged in standard TO 220 packages. The package lead inductances resonated with the device capacitances, which caused severe ringing in the circuit at about 30 MHz (Fig. 13). At an input dc voltage of only 5 V, the drain voltage of the devices rings up to 15 V. The ringing meant that huge currents were flowing through the devices without contributing to the total output power. The efficiency was therefore extremely low (about 35%) and the maximum power output was 30 W. A low-inductance half-bridge package was then designed to reduce the ringing.

The package consists of an etched copper pattern directly bonded to a ceramic substrate $\sim 1\text{--}1/2$ in. long and $3/4$ in. wide (Fig. 14). The substrate could be either alumina or beryllia.

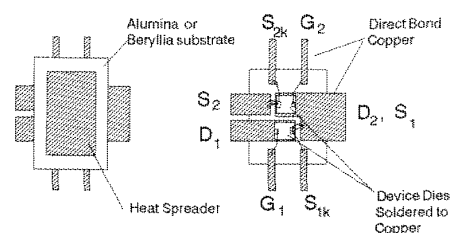


Fig. 14. New low-inductance package.

For this application, alumina was preferred because the cost is lower and it has none of the handling hazards of beryllia. Beryllia could be used for applications where even higher efficiencies are required since it has a much lower thermal impedance than alumina. The package has seven connections to the rest of the circuit: four 50-mil-wide tabs for the two gates and two kelvin sources, each pair of which is located on the small side of the package; two 300-mil tabs on one of the long sides; and finally, a 600-mil-wide tab on the other long side.

One of the 300-mil tabs extends under the die and has the drain of that device soldered to it. The other 300-mil tab is wire-bonded with multiple 5-mil wire bonds to the source of the other device. That device's drain is soldered to an extension of the 600-mil tab, which is also wire-bonded to the source of the first device, again with multiple bonds. Thus the 600-mil tab forms the bridge midpoint, the 300-mil tab connected to the drain of the first device is the dc input voltage connection, and the other 300-mil tab is the ground connection of the half-bridge. The kelvin sources are connected to the corresponding source pad on the die with a single 5-mil wire bond, as are the gate connections.

The printed circuit board is a double-sided board with 2-oz base copper. The ground plane runs under most of the conducting surfaces (Fig. 15). The runs are broad to provide current-carrying capability. A hole is made in the PCB to allow for flat connection of the package tabs to the board. The dc voltage, ground and gate drive connections are made on the bottom of the board. The dc supply is bypassed by several $0.1\text{-}\mu\text{F}$ ceramic disk capacitors which have a self-resonance at close to 13.56 MHz. The midpoint is connected to a broad run on the top of the board. The run is made long enough to clear the heat sink, which is mounted flush to the devices. The Class-E gate drive circuit was designed using the equations developed in [13]. A certain amount of tuning had to be done on the board to account for the parasitics of the layout. The gate drive transformers are coaxial transformers wound on UI 21 H7C1 cores with a 4:1 turns ratio and they are connected in series on the primary side. This is done to allow the use of an IRF 710 MOSFET, which has the lowest gate drive requirement of the Hexfet series. The on resistance of that device, which is equal to 3.5Ω , has very little effect on the efficiency of the gate drive circuit because the reflected load resistance is equal to 32Ω .

A low-inductance probe connection is mounted on the board at the midpoint between the devices to allow measurement of the midpoint voltage to ground. A high-frequency current

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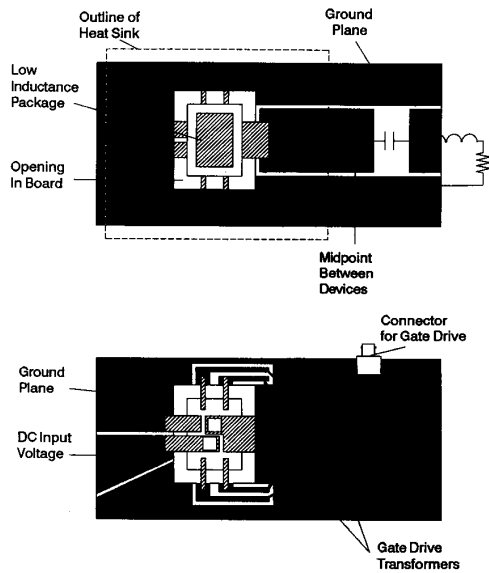


Fig. 15. Printed circuit board layout for Class-D power amplifier using the low-inductance packages.

viewing resistor of $0.1\ \Omega$ is placed in series with the load resistor in the ground return to measure load current. A Tektronix 11402 digital scope is used to measure the voltage and the current and to multiply them. The average of the instantaneous power is then taken as a reading of the power delivered to the load. The scope and probes are calibrated with a standard $50\text{-}\Omega$ water load. The length of the coaxial cable connecting the current viewing resistor to the scope is chosen to give the correct phasing of current versus voltage, as measured by the voltage probe at the operating frequency. The measurements obtained by calorimetry are corrected to account for the resistances that are outside the calorimeter load. This is done by multiplying the power obtained from the thermal measurement by the ratio of the total load resistance with the resistance of the water load. This ratio is equal to 2.5:2.2.

The scope measurements and the calorimetry measurements agree to within 1 W out of 300 W. The thermal measurement is time-consuming to make since one must wait until the load reaches thermal stability, in addition to the constant need for clearing the water lines of bubbles formed in the load as the water is heated. Therefore, since the initial measurements prove the accuracy and reliability of the scope measurements, only these are recorded. However, the thermal measurements are done every so often as a further check. The power measurements also agree with the calculated power budget and the measurement of device heat sink temperature rise.

Fig. 16 shows the midpoint voltage, the load current, and the instantaneous power waveforms of the circuit running off a dc input voltage of 73.1 V with an input current of 4.33 A. The output power is measured to be equal to 298.2 W. The drain efficiency (does not include gate drive) is then equal to 94.2%. The total power dissipated in the gate drive circuit is 9.6 W. Therefore, the overall efficiency is 91.4%. The gate-

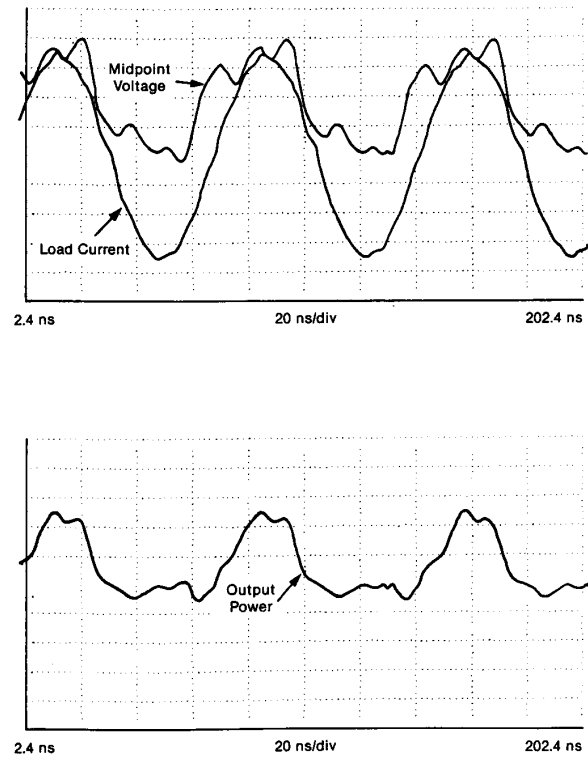


Fig. 16. The midpoint voltage, load current, and instantaneous power waveforms for the Class-D power amplifier operating at an output power of 298 W.

source voltage was 7 V peak, which agrees very well with the value calculated from (33). This drive voltage is relatively low, yet increasing it lowers the efficiency of the circuit. Increasing the gate drive reduces the on-losses and increases the output power but lowers the overall efficiency because of the increased switching losses. Similarly, making the current phase smaller or larger also decreases the efficiency. This is a direct confirmation of the correctness of the analysis performed in this paper.

The nongate drive related power loss in the devices is equal to 18.2 W. The device on-resistances are rated as $85\text{ m}\Omega$ with a 10-V gate drive. At the current levels under consideration and with a 7-V gate drive $R_{ds(on)}$ is approximately the same as with a 10-V drive. The operating temperature of the device is calculated based on a heat sink temperature of 70°C , a junction-to-case thermal resistance of 1 K/W and a case-to-heat-sink thermal resistance of 0.84 K/W to be 96°C (The case is 15.3°C higher than the heat sink and the junction is 11°C higher than the case). Thus, the on-resistance of the device is 1.5 times higher than the 25°C value, which makes it equal to $0.128\ \Omega$. The conduction losses are then equal to 15.15 W for the two devices. Hence, the power losses are accounted for by the gate drive and conduction losses, except for 3 W which could be due to the remaining high-frequency ringing across the devices, or even to some residual switching losses. The power budget indicates that the switching losses are, in fact, very small.

VI. CONCLUSIONS

In this paper, the main loss mechanism in a switching power amplifier at rf is identified as being the discharge of the parasitic output capacitance of the power devices. A method for eliminating it by introducing a dead time between the devices is proposed. A resonant gate drive scheme is introduced that allows fine control of the dead time between the devices at low cost by use of the amplitude of the gate drive. Use of a high-efficiency circuit (such as the Class-E circuit or a low-power Class-D circuit), makes this gate drive scheme very efficient.

The full design equations for very high efficiency Class-D power amplifiers are developed. These design equations satisfy simultaneously the output power requirement and the lossless discharge of the output capacitance of the devices. A low-inductance half-bridge package is designed for operation at rf. This package may be used at lower frequencies where the ringing or the current spikes caused by parasitic inductances are undesirable. Thus, use of such a package in conjunction with a zero-voltage transition resonance switching method should help to reduce the cost of lower frequency switching converters by eliminating the need for snubbers and simplifying the design of EMI filters. An experimental circuit has been built to test the concept. The test circuit is operated at 13.56 MHz with a 300-W output at more than 90% efficiency. This combination of power level, operating frequency, and efficiency is not attainable without the use of zero-voltage switching. Experimental circuits based on the concepts described in this paper have been built and tested at different frequencies and power levels. The agreement between the prediction and the theory in all the circuits was always very good, making this a powerful, robust approach.

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